An Analytic Geodesic from a Computational Basis State to the Dimension-Four SIC Fiducial

We derive the geodesic from $|0\rangle = (1,0,0,0)^T$ to the analytic dimension-four SIC fiducial. The target is returned by `d4_sic_fiducial_ket()` in `sics.py`. We follow the construction in `utils.py`, implemented by `geodesic_interpolator`. Throughout, we use the default cyclic Weyl–Heisenberg basis (`monomial=False`).

The fiducial. Introduce the constants

$$a = \sqrt{2+\sqrt{5}}, \qquad q = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}, \qquad L = \sqrt{2(5+\sqrt{5})}. \tag{1}$$

The function constructs

$$|\phi\rangle = (H \otimes I_2)\,\mathrm{diag}(1,\bar{q},q,i)\frac{(a,1,1,1)^T}{\sqrt{5+\sqrt{5}}} = \frac{1}{L}\begin{pmatrix} a+q \\ \bar{q}+i \\ a-q \\ \bar{q}-i \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{2}$$

In particular, its first component is not real. Write

$$\langle 0|\phi\rangle = \frac{a+q}{L} = re^{i\gamma}, \quad C = |a+q| = \sqrt{a^2+1+\sqrt{2}\,a}, \quad r = \frac{C}{L}, \quad \gamma = \arctan\!\left(\frac{1}{\sqrt{2}\,a+1}\right). \tag{3}$$

Phase alignment and the orthogonal direction. The library first replaces the target by $|\widetilde{\phi}\rangle = e^{-i\gamma}|\phi\rangle$, so that $\langle 0|\widetilde{\phi}\rangle = r > 0$. Define

$$R = \sqrt{L^2 - C^2} = \sqrt{7+\sqrt{5}-\sqrt{2}\,a}, \qquad \omega = \arccos r, \qquad \sin\omega = \frac{R}{L}. \tag{4}$$

The normalized component of the aligned target orthogonal to $|0\rangle$ is

$$|\eta\rangle = \frac{|\widetilde{\phi}\rangle - r|0\rangle}{\sqrt{1-r^2}} = \frac{e^{-i\gamma}}{R}\begin{pmatrix} 0 \\ \bar{q}+i \\ a-q \\ \bar{q}-i \end{pmatrix}. \tag{5}$$

Thus $\langle 0|\eta\rangle = 0$ and $\langle \eta|\eta\rangle = 1$.

The geodesic used by the library. For $0 \le t \le 1$, the normalized horizontal lift of the Fubini–Study geodesic is

$$\boxed{|g(t)\rangle = \cos(t\omega)|0\rangle + \sin(t\omega)|\eta\rangle = \begin{pmatrix} \cos(t\omega) \\ \dfrac{e^{-i\gamma}(\bar{q}+i)}{R}\sin(t\omega) \\ \dfrac{e^{-i\gamma}(a-q)}{R}\sin(t\omega) \\ \dfrac{e^{-i\gamma}(\bar{q}-i)}{R}\sin(t\omega) \end{pmatrix}.} \tag{6}$$

This is the analytic expression returned by `geodesic_interpolator` for these endpoints, apart from numerical roundoff. It satisfies

$$|g(0)\rangle = |0\rangle, \qquad |g(1)\rangle = e^{-i\gamma}|\phi\rangle, \qquad \langle g(t)|g(t)\rangle = 1. \tag{7}$$

The parameter $t\omega$ is linear in t, giving constant speed along the shortest projective geodesic. The curve lies in $\mathrm{span}\{|0\rangle, |\phi\rangle\}$; its intermediate states are not required to be SIC fiducials.

Exact ket endpoints. To require equality to the supplied ket at $t = 1$, including its global phase, choose the phase-adjusted lift

$$\boxed{|f(t)\rangle = e^{it\gamma}|g(t)\rangle = e^{it\gamma}\left[\cos(t\omega)|0\rangle + \frac{e^{-i\gamma}\sin(t\omega)}{R}\big((\bar{q}+i)|1\rangle + (a-q)|2\rangle + (\bar{q}-i)|3\rangle\big)\right].} \tag{8}$$

It obeys exactly

$$|f(0)\rangle = |0\rangle, \qquad |f(1)\rangle = |\phi\rangle, \qquad \langle f(t)|f(t)\rangle = 1. \tag{9}$$

Since $|f(t)\rangle\langle f(t)| = |g(t)\rangle\langle g(t)|$, the two expressions describe the same geodesic of physical states. The additional phase changes the lift in Hilbert space, not the Fubini–Study path.